

Power Amplifier Design

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1 Task Definition and Design Steps

The following report presents a short summary of the design sessions for the course *Design of RF and mm-Wave Integrated Circuits*. The goal of the project was to design a power amplifier using a cascode transistor configuration (in a 45 nm CMOS technology) with input and output matching networks. The design was to be done in such a fashion to achieve a certain 1 dB compression point (*OP1dB* in later text; 22 dB in this case) at a frequency of 24 GHz and with a supply voltage of 2 V. Furthermore, two metrics were to be optimized: Power Added Efficiency (PAE) and the amplifier gain.

The crucial design steps are summarized as follows:

- environment setup (Cadence, ADS) and schematic design, test-bench initialization;
- small-signal (SP) analysis (for input matching network) and large-signal (PSS - load-pull and Harmonic Balance simulations) analysis (for output matching network) - matching done with help of the Smith chart tool in ADS;
- transistor resizing and bias tuning (Cadence) to achieve wanted performance;
- real inductors design (ADS), integration of said components into the final design;
- final simulations of the designed circuit (Cadence).

During the design procedure, several choices were made. To ensure a strong enough cascode structure, transistors were sized fairly large (100 μm per transistor, 20 transistors in parallel) and the input power of 12 dBm was used. Moreover, the bias voltages were swept and tuned to ensure the sufficient OP1dB (hence $V_{B,NM1} = 0.49\text{V}$, $V_{B,NM2} = 1.38\text{V}$). As the circuit exhibited strong unconditional stability, the gate resistance was reduced to $R_G = 1\Omega$. The simulated inductors had a Q-factor of 15, which ensured proper modeling of inductor losses before integrating the real inductor design into the circuit.

However, the design is certainly not ideal, as transistor sizing could have been optimized to ensure reduced area and greater PAE.

2 Schematic of the Final Design

The schematic of the design in question may be seen in Figure 1.

The values of circuit components (transistor lengths and widths, EM simulated inductances, and capacitances) are given in Table 1.

Parameter [unit]	Value
$W_{NM1}[\text{mm}]$	2
$W_{NM2}[\text{mm}]$	2
$L_{NM1}[\text{nm}]$	45
$L_{NM2}[\text{nm}]$	45
$L_{in}[\text{pH}]$	37.65
$L_{in2}[\text{pH}]$	143.37
$L_{out}[\text{pH}]$	37.65
$L_{out2}[\text{pH}]$	88.82
$C_{in}[\text{fF}]$	434.52
$C_{out}[\text{fF}]$	448.35

Table 1: Power amplifier components' values.

3 Simulation of the Final Design

Simulation results of the final design are given in Figures 2 - 9. The modeled inductors' layouts and dimensions are given in Figure 10.

4 Final Amplifier Performance

The final values of the amplifier performance parameters can be seen in Table 2.

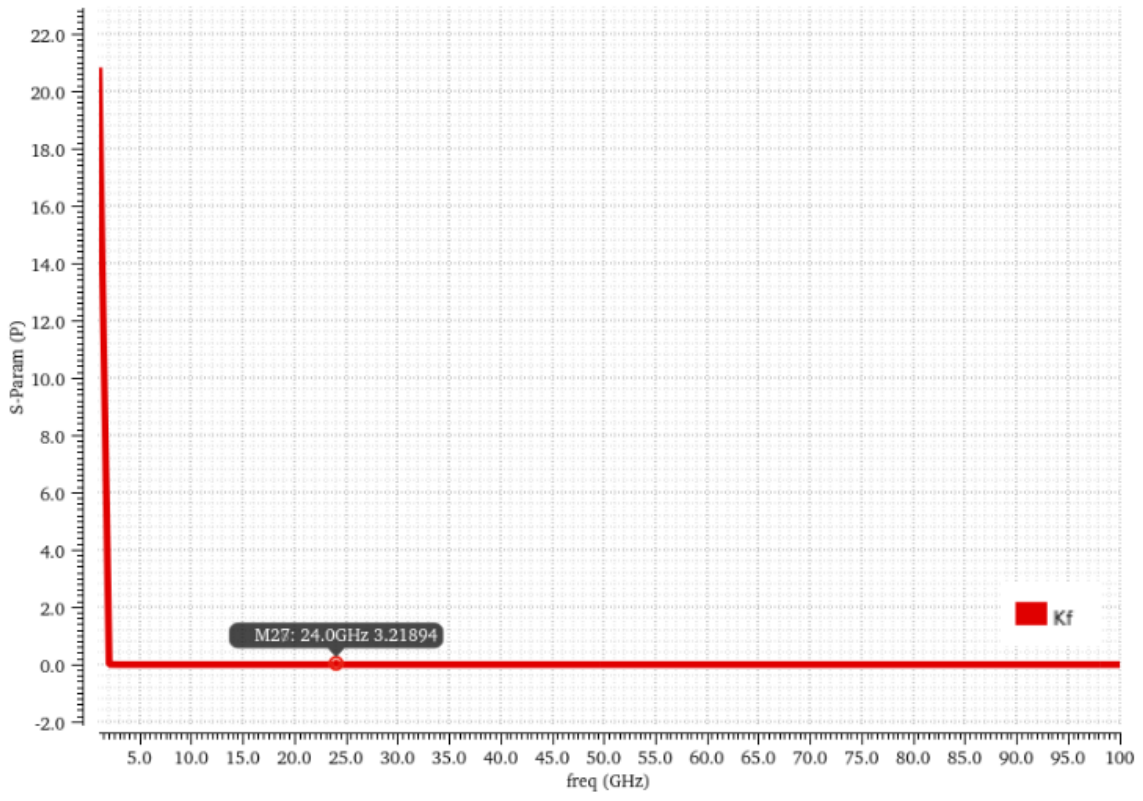


Figure 2: Simulated results for K_f .

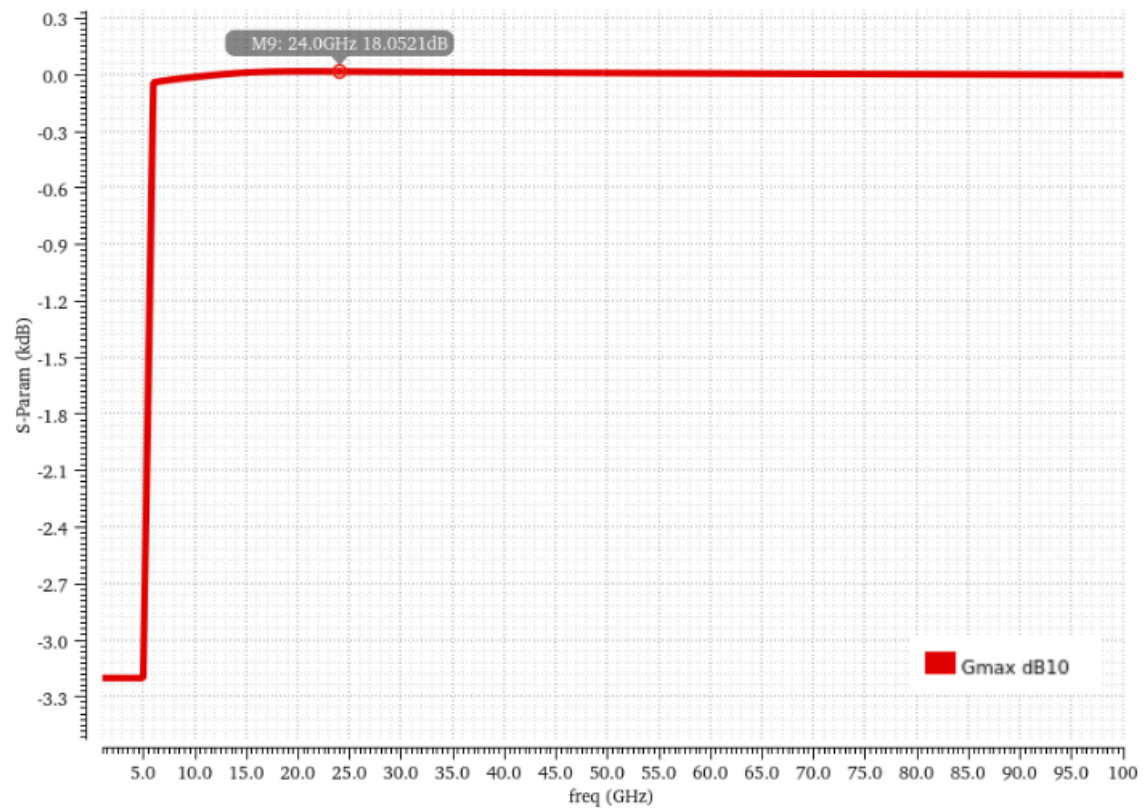


Figure 3: Simulated results for G_{max} .

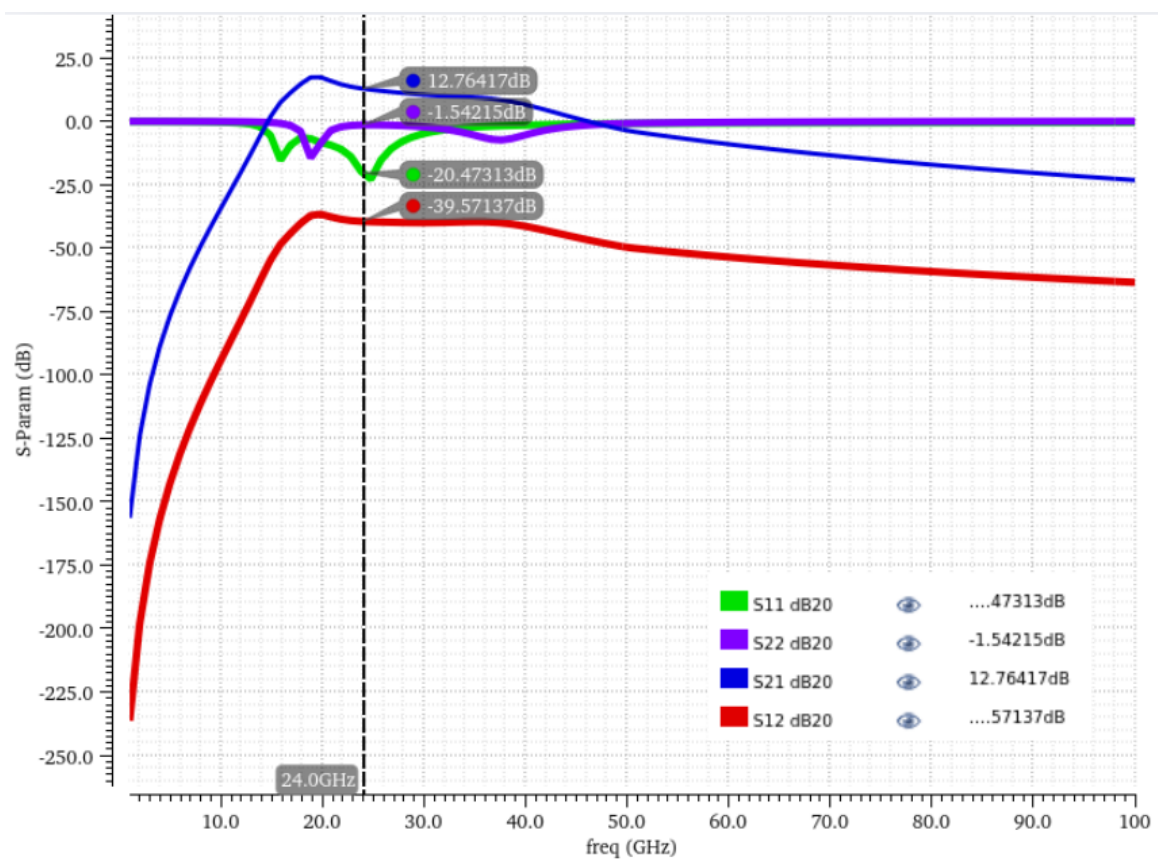


Figure 4: Simulated results for S-parameters.

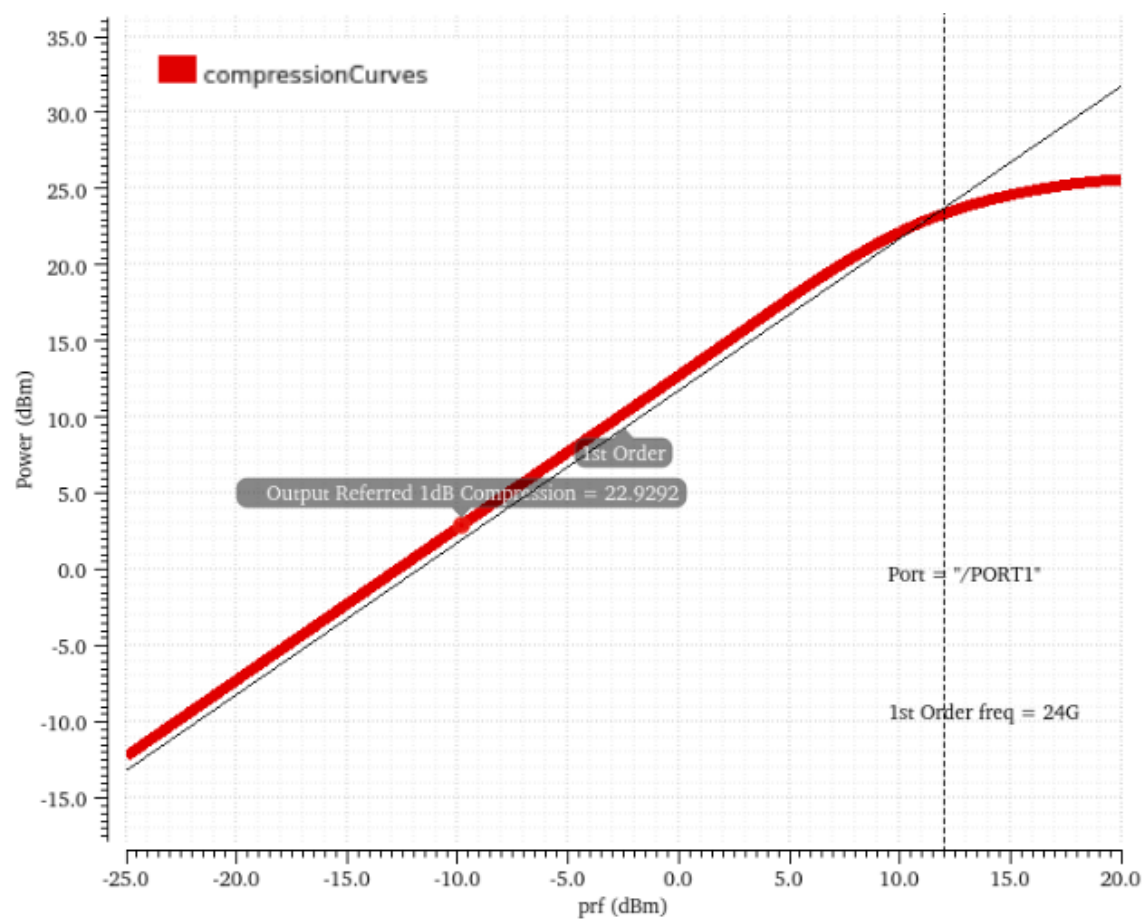


Figure 5: Simulated results for OP_{1dB} .

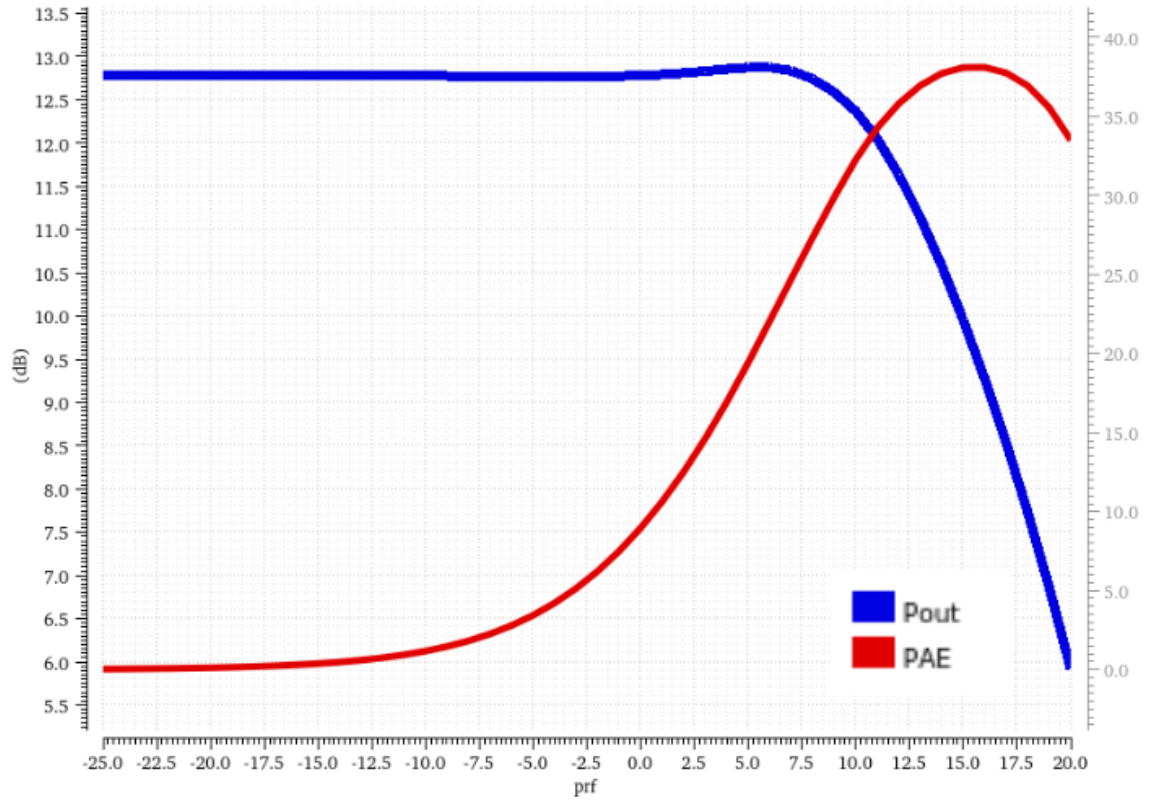


Figure 6: Simulated results for PAE and output power gain (P_{out}).

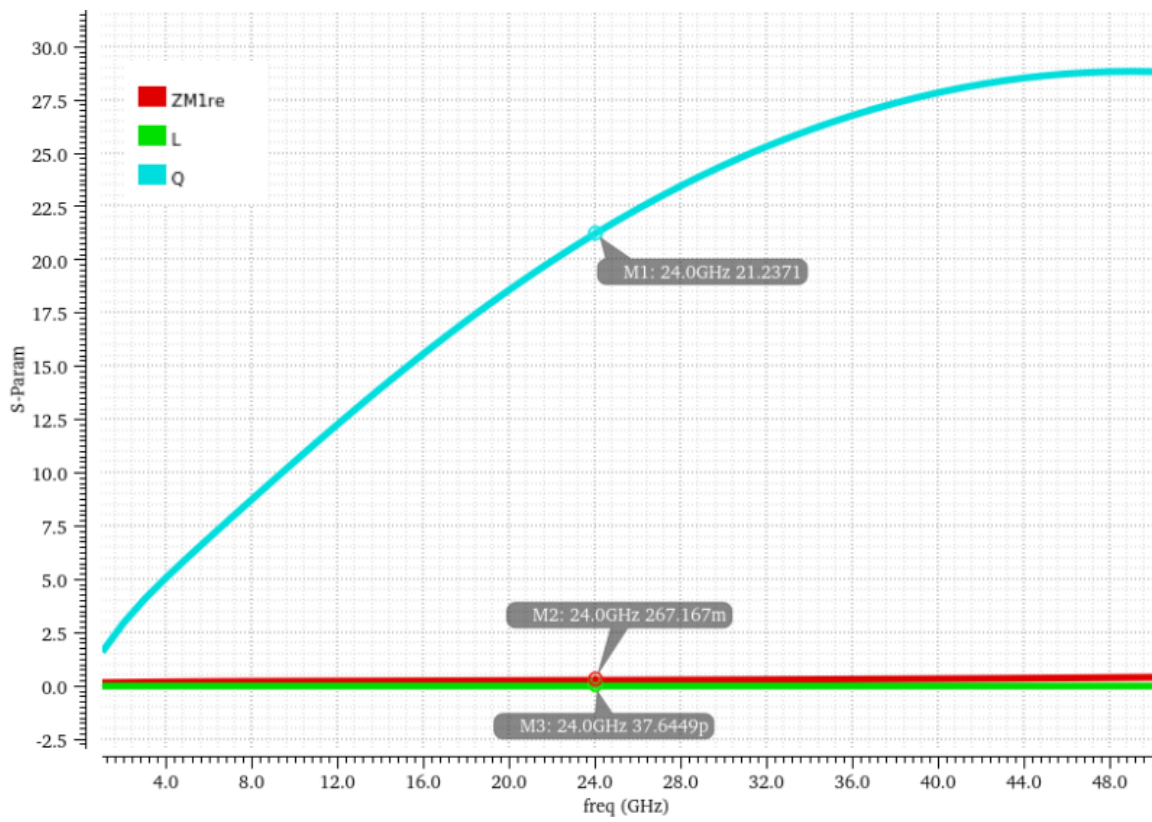


Figure 7: Simulated results for L_{in} and L_{out} .

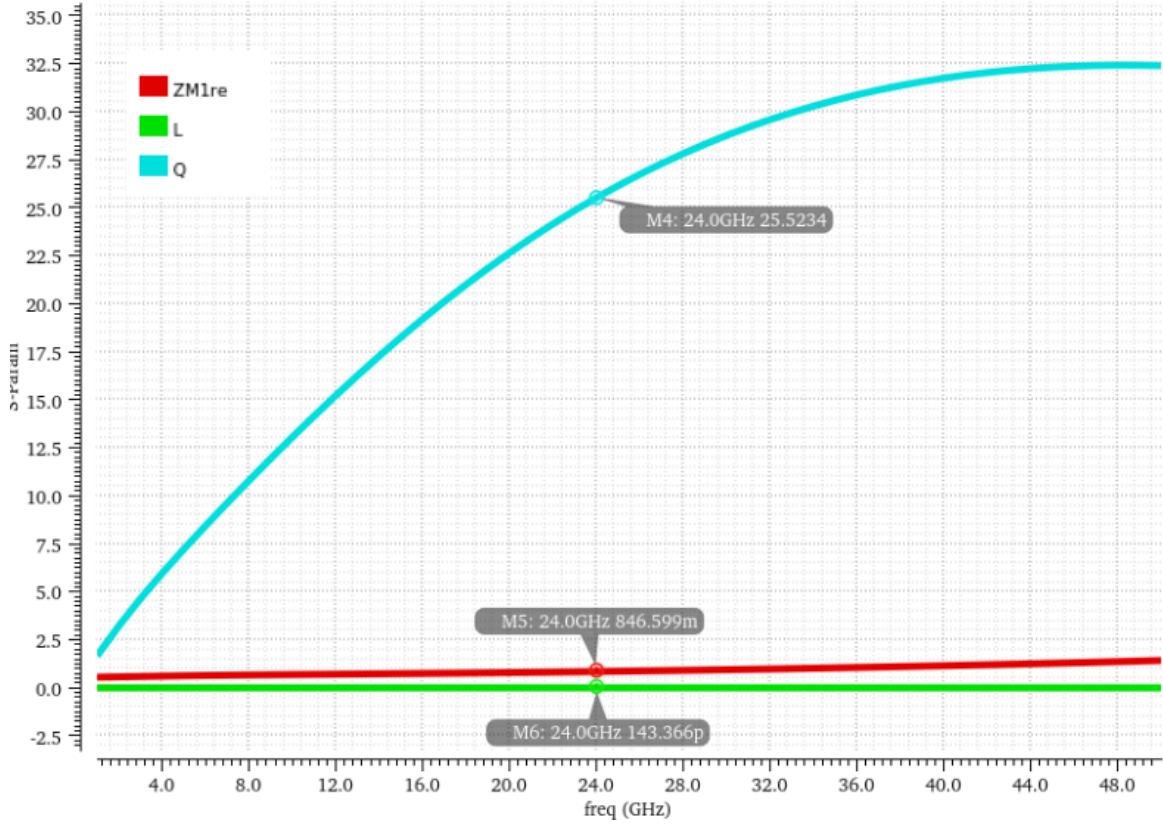


Figure 8: Simulated results for L_{in2} .

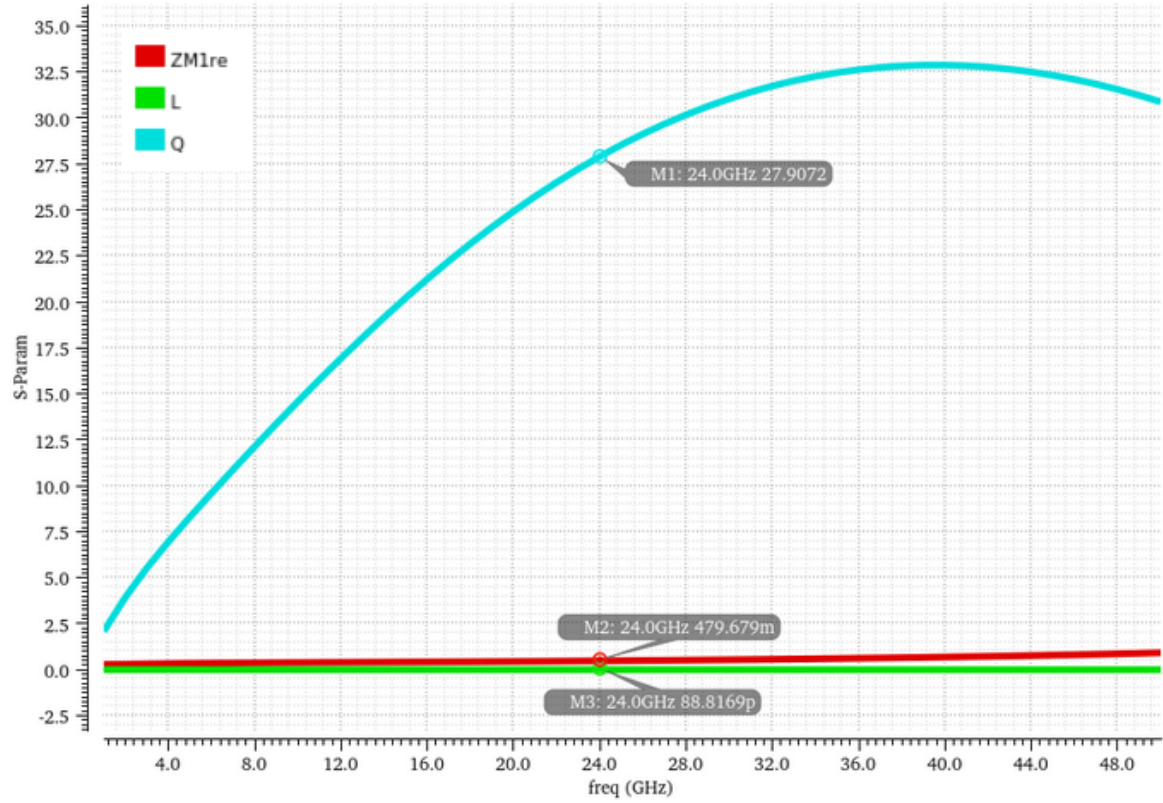


Figure 9: Simulated results for L_{out2} .

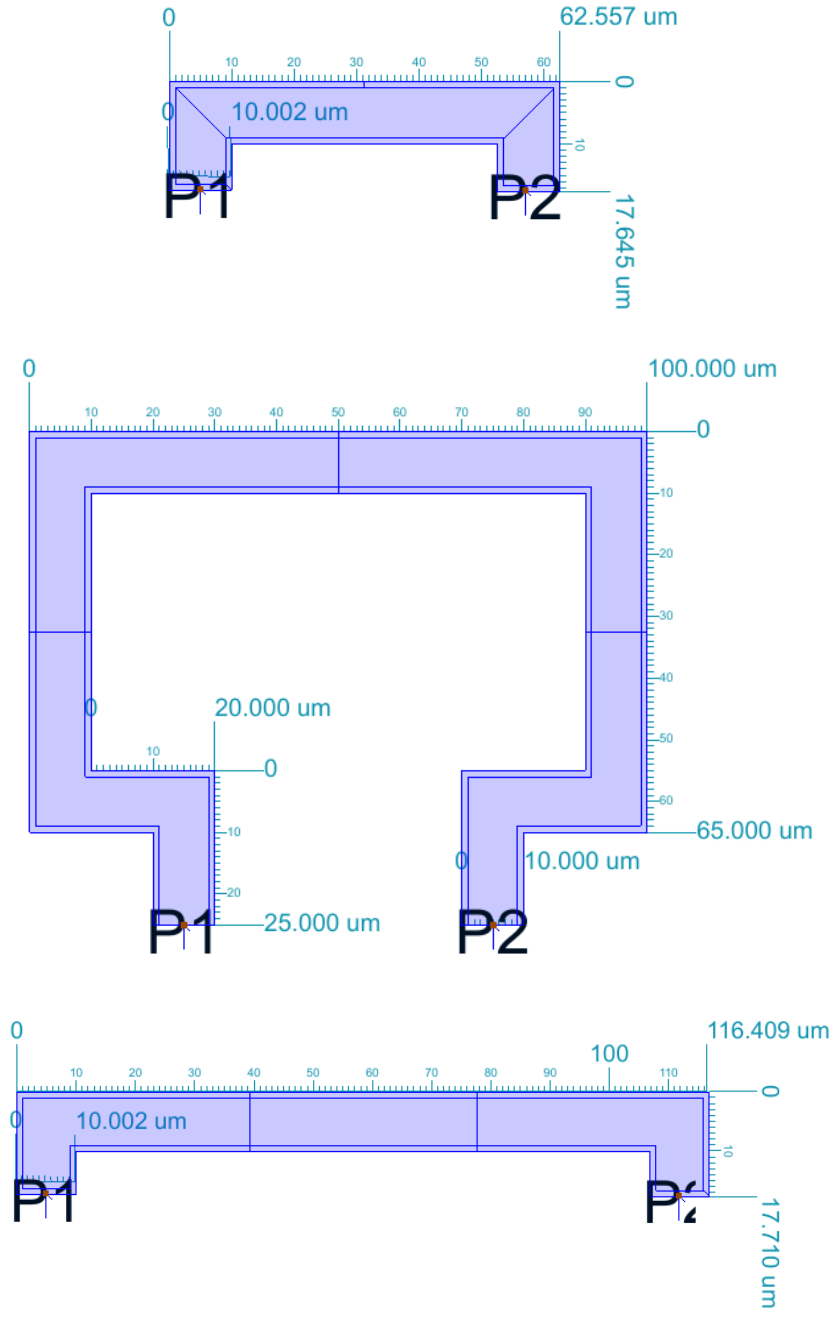


Figure 10: The real inductors' layouts with dimensions ($L_{in,out}$, L_{in2} , L_{out2} , respectively).